

## Design Patterns:Module-01

### Logger file:

```
public class Logger {  
  
    private static Logger instance;  
  
    private Logger() {  
    }  
  
    public static Logger getInstance() {  
        if (instance == null) {  
            instance = new Logger();  
        }  
        return instance;  
    }  
  
    public void log(String message) {  
        System.out.println("LOG: " + message);  
    }  
}
```

### Testlogger file:

```
public class TestLogger {  
  
    public static void main(String[] args) {  
  
        Logger logger1 = Logger.getInstance();  
        Logger logger2 = Logger.getInstance();  
  
        logger1.log("Application Started");  
        logger2.log("Logging Information");  
  
        if (logger1 == logger2) {  
            System.out.println("Only one instance of Logger is created.");  
        } else {  
            System.out.println("Multiple instances are created.");  
        }  
    }  
}
```

### Output:

```

Windows PowerShell
At line:1 char:3
+ & "C:\Program Files\Java\jdk-22.0.2\bin\java.exe" -version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Program File....2\bin\java.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample> Get-ChildItem "C:\Program Files\Java"

Directory: C:\Program Files\Java

Mode                LastWriteTime         Length Name
----                -
d-----          14-06-2026   19:01             jdk-22

PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample> & "C:\Program Files\Java\jdk-22\bin\java.exe" -version
java version "22.0.2" 2024-07-16
Java(TM) SE Runtime Environment (build 22.0.2+9-70)
Java HotSpot(TM) 64-Bit Server VM (build 22.0.2+9-70, mixed mode, sharing)
PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample> & "C:\Program Files\Java\jdk-22\bin\javac.exe" -version
javac 22.0.2
PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample> & "C:\Program Files\Java\jdk-22\bin\javac.exe" Logger.java TestLogger.java
PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample> & "C:\Program Files\Java\jdk-22\bin\java.exe" TestLogger
LOG: Application Started
LOG: Logging Information
Only one instance of Logger is created.
PS C:\Users\vanka\OneDrive\Desktop\SingletonPatternExample>

```

## Exercise 2: Implementing the Factory Method Pattern

### File 1: Document.java

```

public interface Document {
    void open();
}

```

### File 2: WordDocument.java

```

public class WordDocument implements Document {

```

```

    public void open() {
        System.out.println("Opening Word Document");
    }
}

```

### File 3: PdfDocument.java

```

public class PdfDocument implements Document {

```

```

    public void open() {
        System.out.println("Opening PDF Document");
    }
}

```

### File 4: ExcelDocument.java

```

public class ExcelDocument implements Document {

```

```

    public void open() {
        System.out.println("Opening Excel Document");
    }
}

```

### File 5: DocumentFactory.java

```

public abstract class DocumentFactory {
    public abstract Document createDocument();
}

```

### File 6: WordDocumentFactory.java

```

public class WordDocumentFactory extends DocumentFactory {

```

```

    public Document createDocument() {
        return new WordDocument();
    }
}

```

File 7: PdfDocumentFactory.java

```
public class PdfDocumentFactory extends DocumentFactory {  
  
    public Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

File 8: ExcelDocumentFactory.java

```
public class ExcelDocumentFactory extends DocumentFactory {  
  
    public Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

**File 9: TestFactoryMethod.java**

```
public class TestFactoryMethod {  
  
    public static void main(String[] args) {  
  
        DocumentFactory wordFactory = new WordDocumentFactory();  
        Document word = wordFactory.createDocument();  
        word.open();  
  
        DocumentFactory pdfFactory = new PdfDocumentFactory();  
        Document pdf = pdfFactory.createDocument();  
        pdf.open();  
  
        DocumentFactory excelFactory = new ExcelDocumentFactory();  
        Document excel = excelFactory.createDocument();  
        excel.open();  
    }  
}
```

